BRIDGE-DN — Build Diagrams

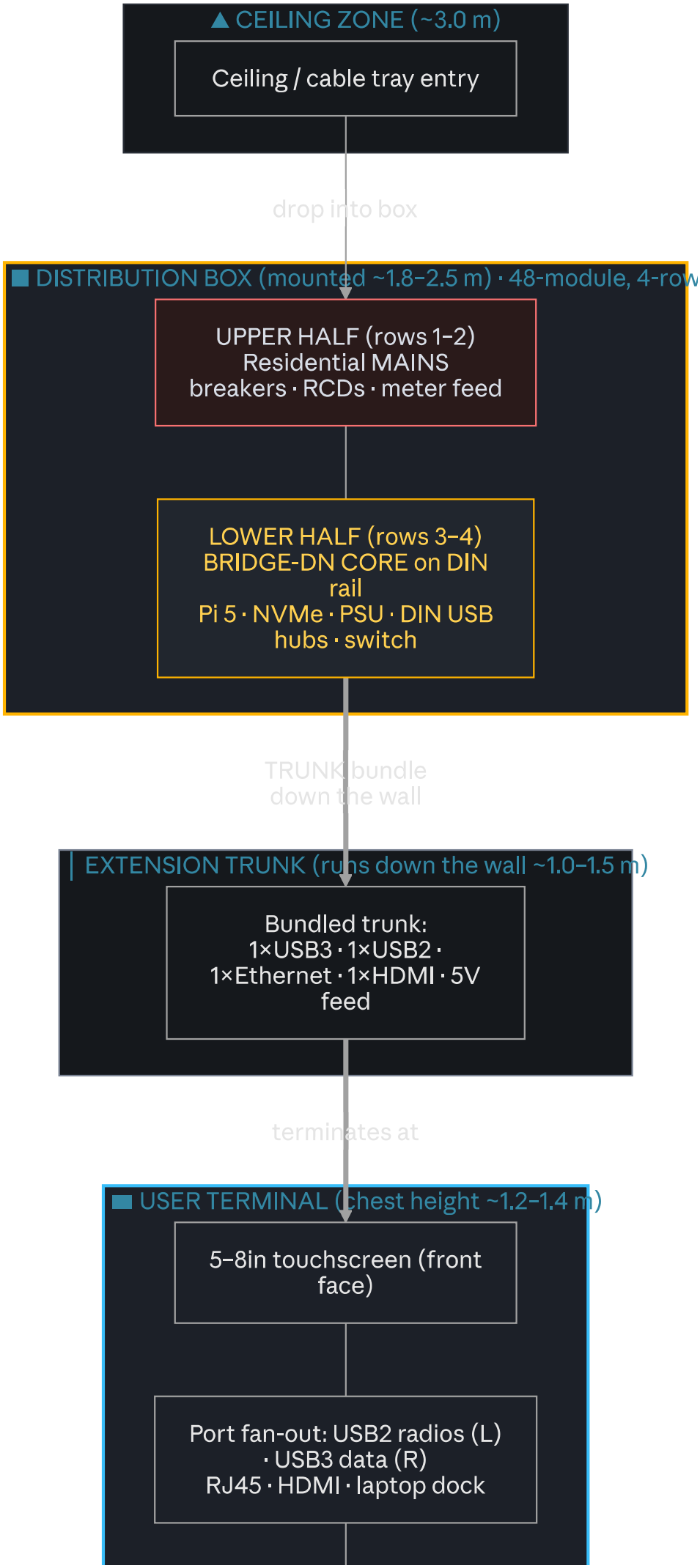
Visual reference for the BRIDGE-DN installation. Diagrams use **vertical position to represent actual height on the wall** — top of a diagram = high on the wall, bottom = chest/hand height. Arrows show cable/signal direction and physical routing.

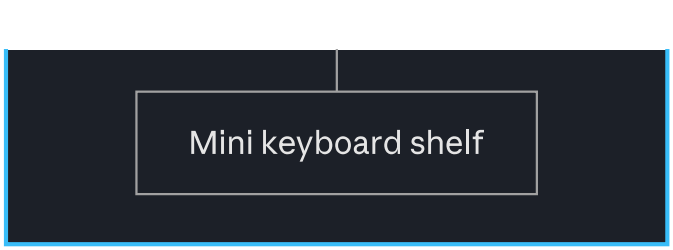
Apartment context: Király Street, Budapest · ceiling > 3 m · distribution box mounted high under the mains · Pi core at ~2.5 m · user terminal at chest level (~1.2–1.4 m).

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1. Wall Elevation — where everything physically sits

This is the "where on the wall" master view. Read top-to-bottom as high-to-low on the wall.


▲ CEILING ZONE (~3.0 m)
Ceiling / cable tray entry
drop into box
DISTRIBUTION BOX (mounted ~1.8–2.5 m) · 48-module, 4-row
UPPER HALF (rows 1–2)
Residential MAINS
breakers · RCDs · meter feed
LOWER HALF (rows 3–4)
BRIDGE-DN CORE on DIN rail
Pi 5 · NVMe · PSU · DIN USB hubs · switch
TRUNK bundle down the wall
EXTENSION TRUNK (runs down the wall ~1.0–1.5 m)
Bundled trunk:
1×USB3 · 1×USB2 · 1×Ethernet · 1×HDMI · 5V feed
terminates at
USER TERMINAL (chest height ~1.2–1.4 m)
5–8in touchscreen (front face)
Port fan-out: USB2 radios (L) · USB3 data (R)
RJ45 · HDMI · laptop dock

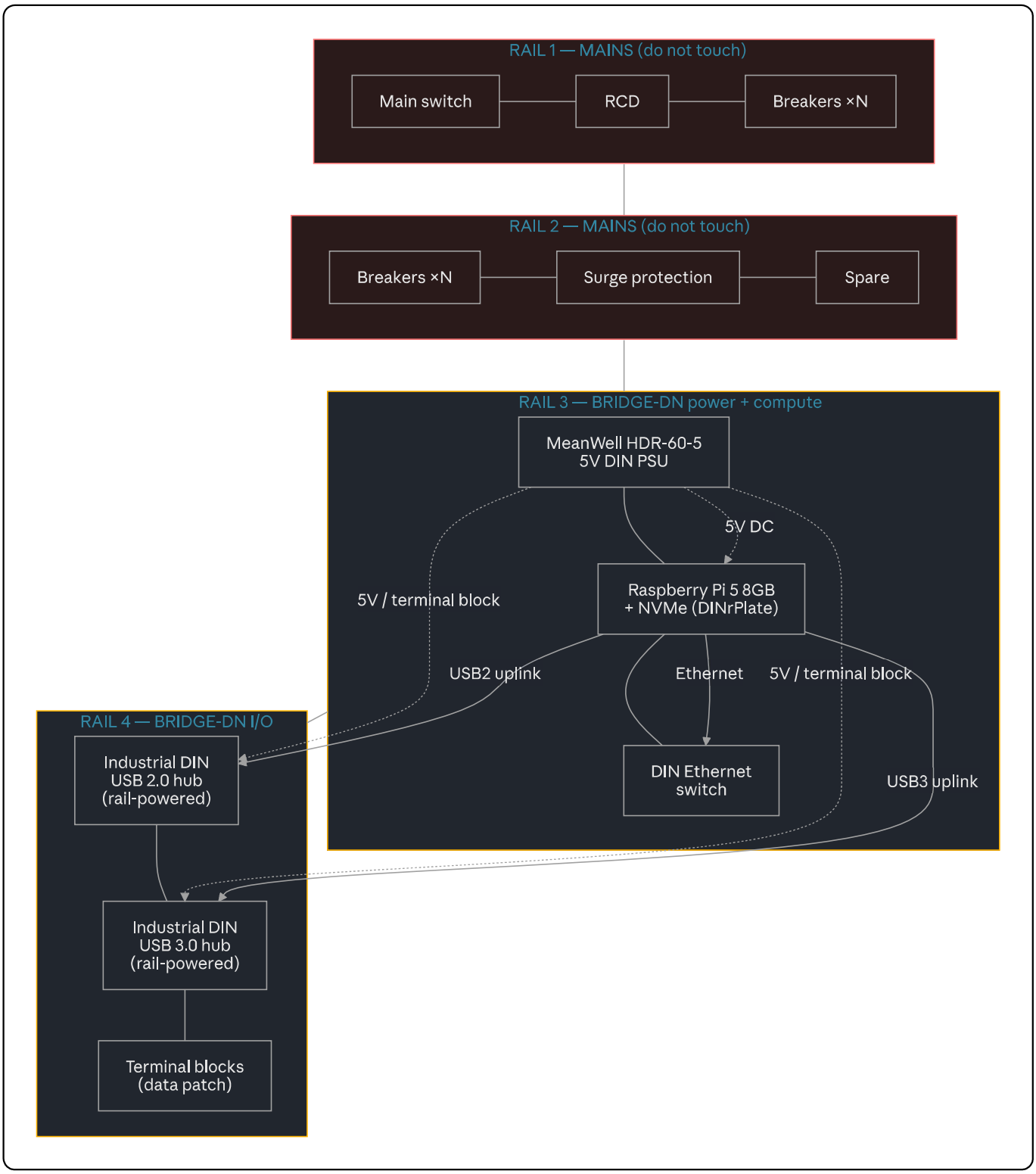


Placement key: MAINS occupies the upper half (untouched, live 230V). BRIDGE-DN CORE occupies the free lower half. The trunk drops straight down the wall to the terminal at chest height so a person can plug in without a ladder.

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2. Distribution Box Internals — DIN rail layout

Zoom into the box. Each row is a 12-module DIN rail. Top-to-bottom = top rail to bottom rail.

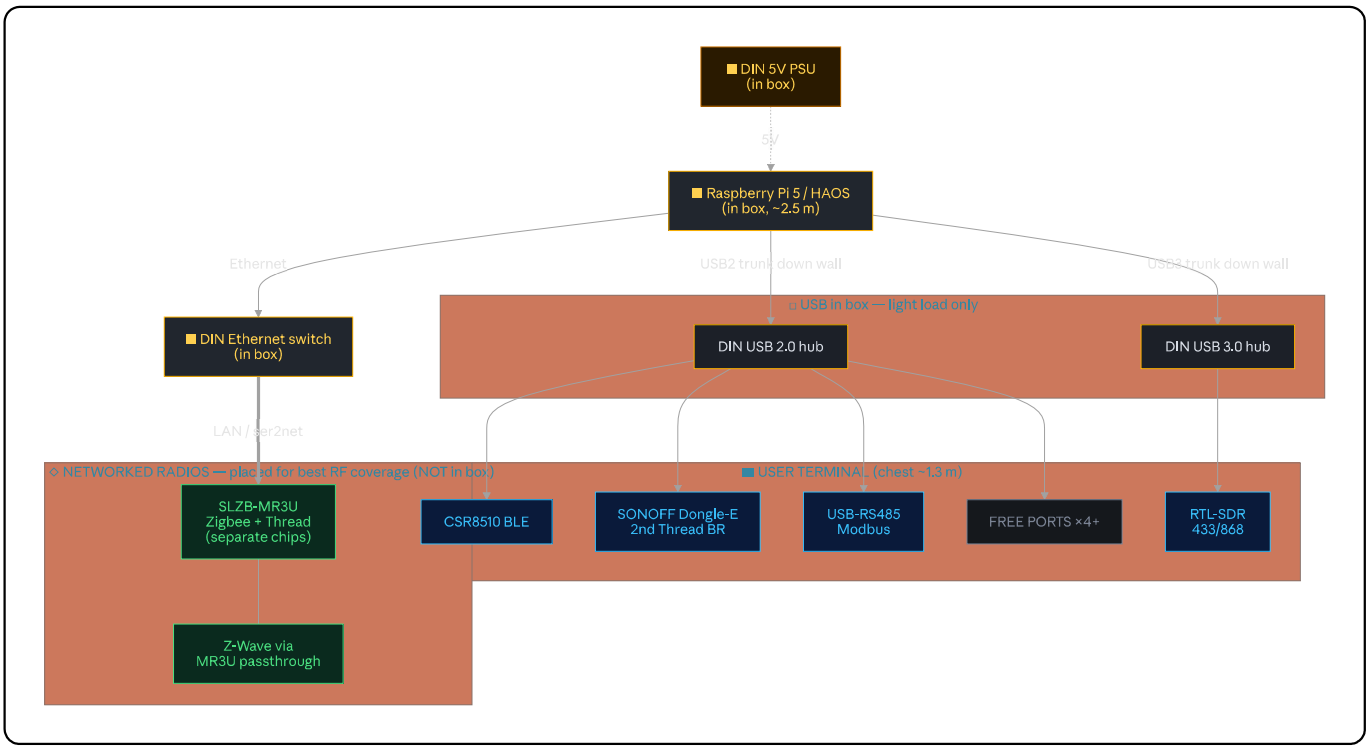


Note: The 230V mains half and the 5V DC BRIDGE half share the box but are physically separated by rail. Only the lower two rails carry the smart-home core.

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3. BRIDGE-DNE (LAN version) — signal architecture

Heavy radios ride the LAN (ser2net), placed away from the box. Green = networked, amber = USB, blue = terminal.

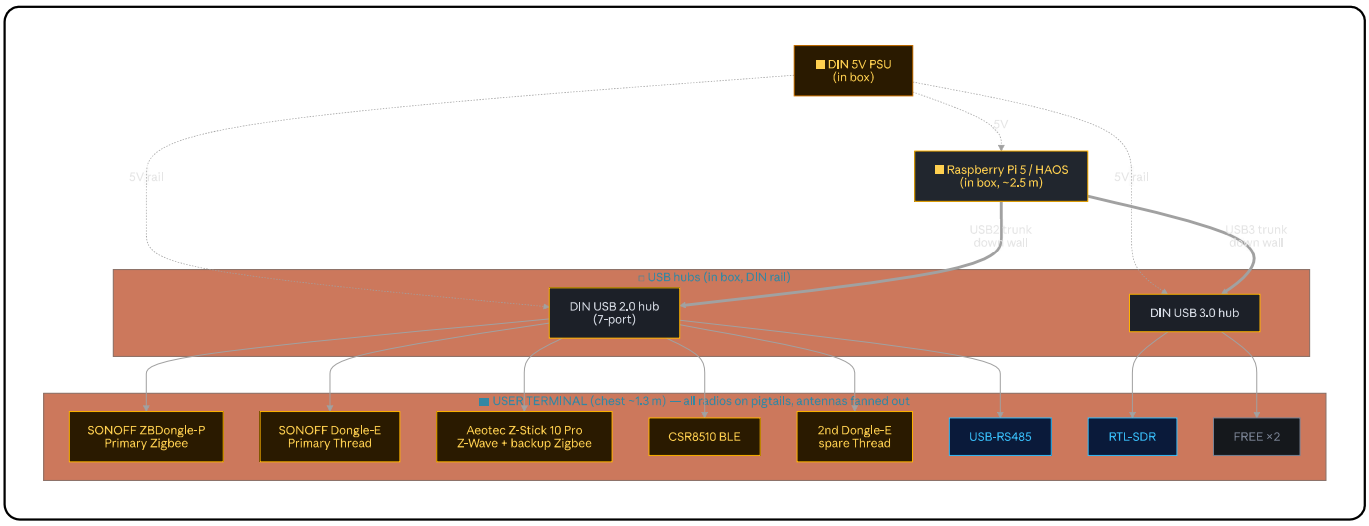


DNE placement logic: the SLZB-MR3U (Zigbee + Thread) leaves the box entirely and sits wherever coverage is best. Only light, non-RF-critical USB devices (BLE, spare Thread, RS-485, SDR) hang at the terminal. ~2 radio USB ports used.

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4. BRIDGE-DNU (USB version) — signal architecture

Everything is USB. No LAN dependency. All radios hang at the terminal on extension pigtails.



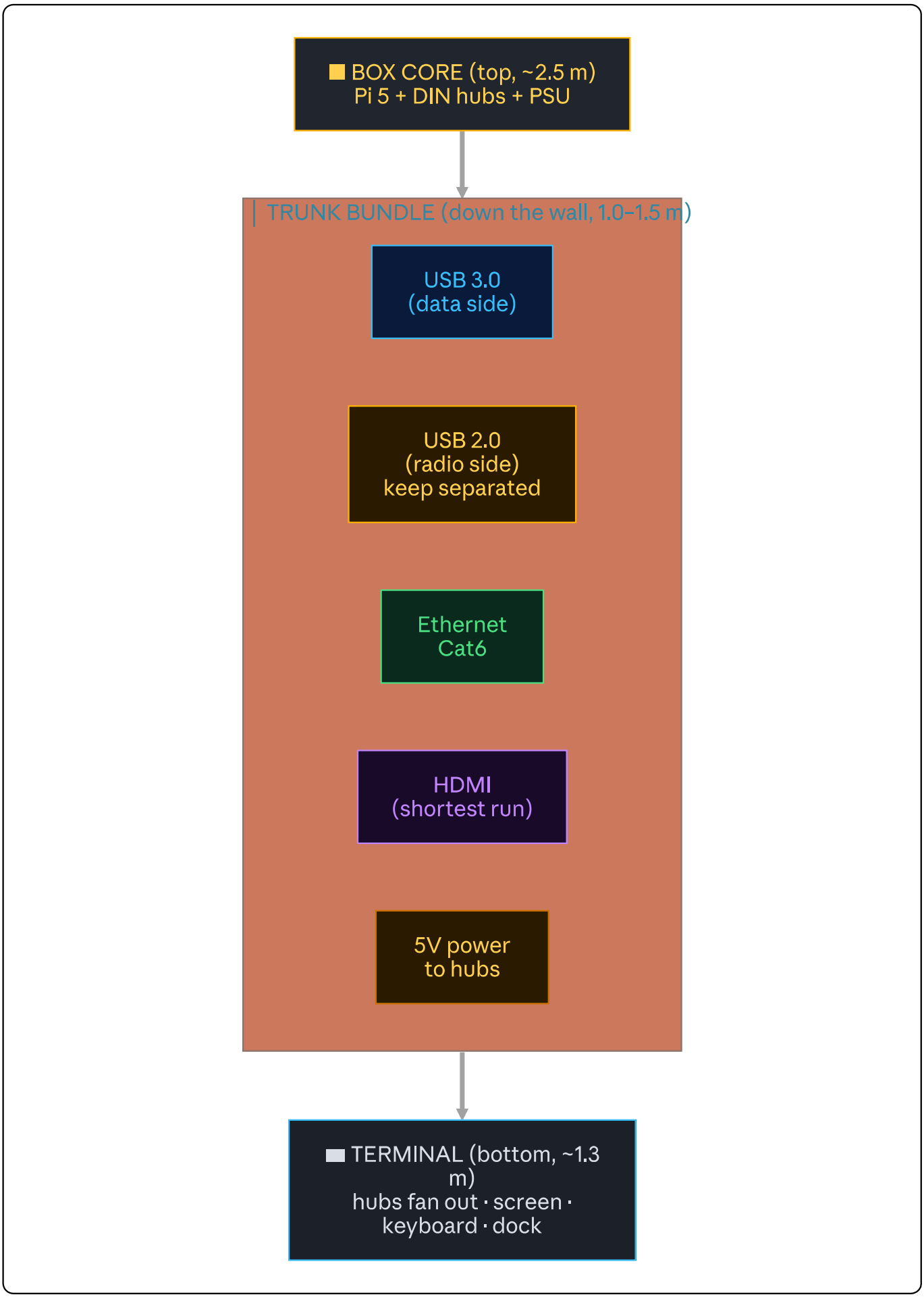
DNU placement logic: no networked radios — every 2.4 GHz dongle plugs into the terminal's USB 2.0 hub via a 15–30 cm pigtail so antennas sit clear of the metal box and each other. ~5–6 radio USB ports used, zero network dependency.

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5. The Extension Trunk — what runs down the wall

Detail of the bundle between box and terminal. Left-to-right = the cables side by side in the

trunk.



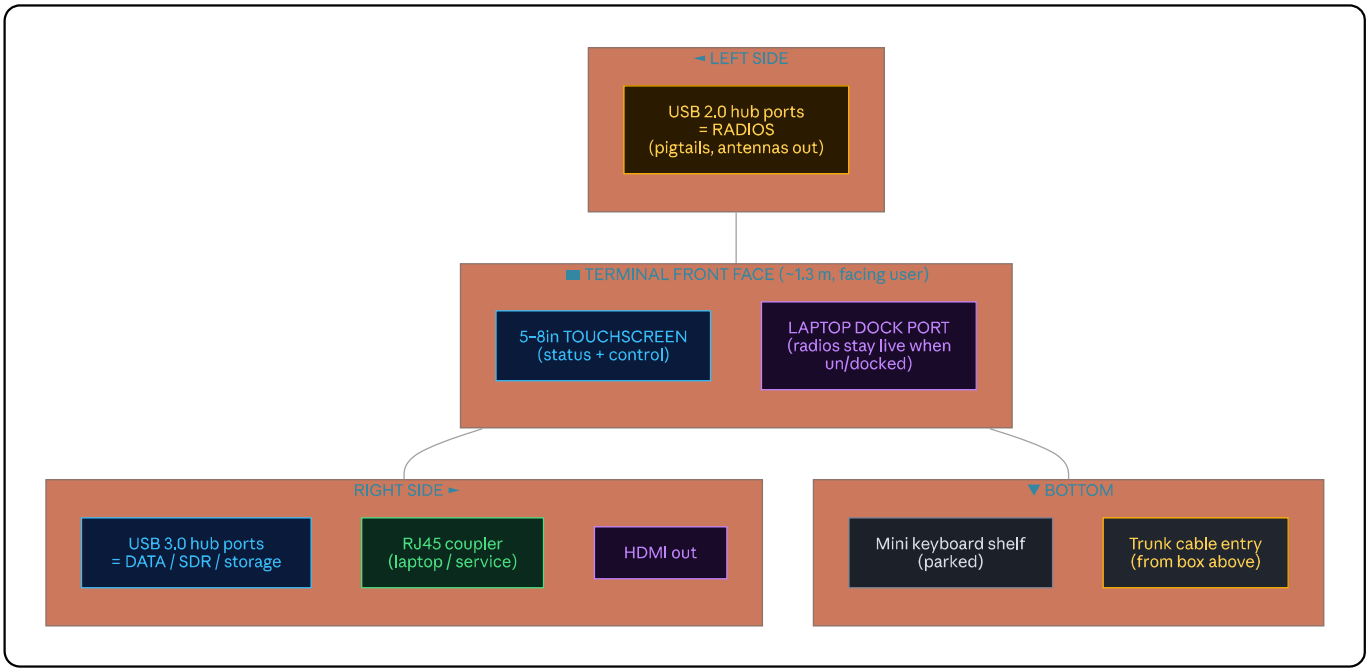
Trunk rule: keep the USB 2.0 (radio) line physically separated from the USB 3.0 line inside the bundle — USB 3.0 broadband noise couples into 2.4 GHz radios. Run them on opposite

sides, or use foil-shielded USB 3.0 cable. Power (USB-C to Pi) is NOT extended — it stays in the box.

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6. Terminal Face — user-level port layout

Front view of the chest-level terminal (Version C/D style: screen + port block + dock).



Face logic: radios on the LEFT (USB 2.0, RF-isolated, antennas pointing away from wall), data on the RIGHT (USB 3.0), trunk enters from BOTTOM (coming down from the box above), keyboard parks underneath. Laptop dock on the front keeps radios connected to the Pi whether or not a laptop is present.

Legend

Colour	Meaning
Red border	230V mains — do not touch
Amber border	BRIDGE-DN core / USB / power (5V DC)
Green border	Networked radios (LAN / ser2net) — DNE only
Blue border	User terminal / data-side devices
Purple border	Video (HDMI) / laptop dock
Grey border	Spare / free / passive routing

Height reference across all diagrams: ceiling ~3.0 m → distribution box 1.8–2.5 m → trunk down the wall → user terminal 1.2–1.4 m.